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AWS-DEVOPS



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Why Users & Groups Matter



Linux servers are shared environments

- Users control access and ownership
- Groups simplify permission management
- Common DevOps users: developers, admins, docker, CI/CD users

Users define identity. Groups make access management easier to scale across a team.

Basic User Commands



Basic User Commands



sudo Command

- The sudo (superuser do) command grants non-administrative users certain administrative privileges.
- When you place sudo before any command in the terminal, that command runs with elevated privileges.
- If you're unsure whether you're using sudo or su, check the last character of the command prompt. If it's a pound sign (#), it means you've logged in as root.

Basic User Commands



sudo Command

Commands	Meaning
sudo -l	List available commands.
sudo command	Run command as root.
sudo -u root command	Run command as root.
sudo -u user command	Run command as user.
sudo su	Switch to the superuser account.
sudo su -	Switch to the superuser account with root's environment.
sudo su - username	Switch to the username's account with the username's environment.
sudo -s	Start a shell as root
sudo -u root -s	Same as above.
sudo -u user -s	Start a shell as user.

sudo & Privilege Management



sudo runs a single command with elevated privileges

```
$ sudo command  
$ sudo su -  
$ sudo visudo
```

- Prefer: sudo command
- Avoid working permanently as root
- Use visudo for safe sudo configuration changes

sudo su - creates a root login shell when required.

Basic User Commands



whoami	current user
hostname	shows the system hostname
hostname -i	Show the IP address of the system

```
linuxuser@techproeducation: ~$ whoami
linuxuser
linuxuser@techproeducation: ~$ hostname
techproeducation
linuxuser@techproeducation: ~$
```

Basic User Commands



Whatis	display one-line manual page names.
Apropos	search the manual page names and descriptions.

```
linuxuser@techproeducation: ~$ apropos pwd
pwd (1) - print name of current/working directory
pwdx (1) - report current working directory of a process
unix_chkpwd (8) - Helper binary that verifies the password of the current user
linuxuser@techproeducation: ~$ whatis pwd
pwd (1) - print name of current/working directory
linuxuser@techproeducation: ~$
```

Basic User Commands



who provide with details about who is logged on the system.

```
linuxuser@techproeducation: ~$ who
root      pts/1        2023-12-02 22:19
linuxuser@techproeducation: ~$ |
```

Basic User Commands



w inform who is logged on and what they are doing.

```
linuxuser@techproeducation: ~$ w
 23:11:49 up 52 min,  1 user,  load average: 0.00, 0.00, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
root      pts/1    -                22:19   51:54  0.02s  0.01s  -bash
linuxuser@techproeducation: ~$ |
```

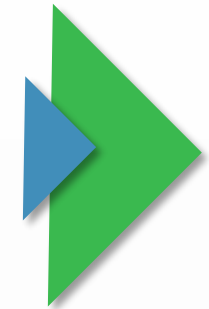
Basic User Commands



id Display user id, your primary group id, and a list of the groups you belong to.

id [username]

```
linuxuser@techproeducation: ~$ id
uid=1000(linuxuser) gid=1000(linuxuser) groups=1000(linuxuser)
linuxuser@techproeducation: ~$ |
```



User management



User management



etc/passwd On Linux, the local user database is /etc/passwd.

```
linuxuser@techproeducation: ~$ tail -3 /etc/passwd
uidd:x:106:112::/run/uidd:/usr/sbin/nologin
tcpdump:x:107:113::/nonexistent:/usr/sbin/nologin
linuxuser:x:1000:1000::/home/linuxuser:/bin/bash
linuxuser@techproeducation: ~$
```

Understanding /etc/passwd



Example

```
ec2-user:x:1000:1000:EC2 User:/home/ec2-user:/bin/bash
```

Fields

username | password placeholder | UID | GID | description | home | shell

User management



useradd useradd command is used for creating a new user.
useradd [username]

```
root@techproeducation: ~  
root@techproeducation:~# useradd -m -d /home/techpro -c "user techpro" techpro  
root@techproeducation:~# tail -1 /etc/passwd  
techpro:x:1001:1001:user techpro:/home/techpro:/bin/sh  
root@techproeducation:~# |
```

- -m is used to force the creation of the home directory.
- -d is used to set the name of the home directory.
- -c is used to set a description.

User management



adduser

adduser is not a standard Linux command. It's basically a Perl script that uses the useradd command in the background. This is more effective at creating new users on Linux.

```
root@techproeducation: ~  
root@techproeducation:~# adduser sam  
Adding user 'sam' ...  
Adding new group 'sam' (1003) ...  
Adding new user 'sam' (1003) with group 'sam' ...  
Creating home directory '/home/sam' ...  
Copying files from '/etc/skel' ...  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for sam  
Enter the new value, or press ENTER for the default  
  Full Name []: sam dai  
  Room Number []: 1  
  Work Phone []: 1  
  Home Phone []: 1  
  Other []: 1  
Is the information correct? [Y/n] y  
root@techproeducation:~# tail -1 /etc/passwd  
sam:x:1003:1003:sam dai,1,1,1,1:/home/sam:/bin/bash  
root@techproeducation:~#
```

User management



userdel

Delete a user. userdel command will not remove the user's home directory from the file system. If you want to remove the home directory, you need to use the -r in the command line.

userdel [username]

```
root@techproeducation: ~  
root@techproeducation:~# userdel -r sam  
userdel: sam mail spool (/var/mail/sam) not found  
root@techproeducation:~# tail -1 /etc/passwd  
mauro:x:1002:1002:Mauro,,,:/home/mauro:/bin/bash  
root@techproeducation:~# userdel -r mauro  
userdel: mauro mail spool (/var/mail/mauro) not found  
root@techproeducation:~# tail -1 /etc/passwd  
techpro:x:1001:1001:user techpro:/home/techpro:/bin/sh  
root@techproeducation:~# userdel -r techpro  
userdel: techpro mail spool (/var/mail/techpro) not found  
root@techproeducation:~# tail -1 /etc/passwd  
linuxuser:x:1000:1000::/home/linuxuser:/bin/bash  
root@techproeducation:~#
```

User management

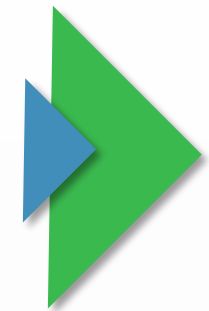


usermod

Modify a user's properties. The example below modifies the description of the user linuxuser.

usermod -[option] [value] [username]

```
root@techproeducation: ~  
root@techproeducation:~# tail -1 /etc/passwd  
linuxuser:x:1000:1000:~/home/linuxuser:/bin/bash  
root@techproeducation:~# usermod -c "linux admin" linuxuser  
root@techproeducation:~# tail -1 /etc/passwd  
linuxuser:x:1000:1000:linux admin:/home/linuxuser:/bin/bash  
root@techproeducation:~# |
```



User Passwords



User Passwords



passwd User passwords can be set with the passwd command.

passwd [username]

```
root@techproeducation: ~  
root@techproeducation:~# passwd linuxuser  
New password:  
Retype new password:  
passwd: password updated successfully  
root@techproeducation:~#
```

User Passwords



etc/shadow

User passwords are encrypted and stored in /etc/shadow file. The /etc/shadow file is only read and can be accessed by root only.

```
root@techproeducation: ~  
root@techproeducation:~# tail -1 /etc/shadow  
linuxuser:$y$j9T$6F8Sv.qx.yglVHBL0DezJ.$v1vDs5zFHDWdWuJySukN  
Y79MWGd4rEx0UIVsod.FS.D:19693:0:99999:7:::  
root@techproeducation:~#
```

User Passwords



etc/login.defs

The /etc/login.defs file includes some default user password settings, such as password aging and length settings.

```
root@techproeducation: ~  
root@techproeducation:~# grep "PASS" /etc/login.defs  
# PASS_MAX_DAYS Maximum number of days a password may be used.  
# PASS_MIN_DAYS Minimum number of days allowed between password changes.  
# PASS_WARN_AGE Number of days warning given before a password expires.  
PASS_MAX_DAYS 99999  
PASS_MIN_DAYS 0  
PASS_WARN_AGE 7  
#PASS_CHANGE_TRIES  
#PASS_ALWAYS_WARN  
#PASS_MIN_LEN  
#PASS_MAX_LEN  
# NO_PASSWORD_CONSOLE  
root@techproeducation:~#
```

Password Aging with chage



- Passwords are stored as cryptographic hashes in /etc/shadow
- /etc/shadow is accessible only by root

Commands

```
$ chage -l username  
$ chage -M 90 username  
$ chage -d 0 username
```

EC2 SSH Key Authentication



EC2 uses key-based authentication.

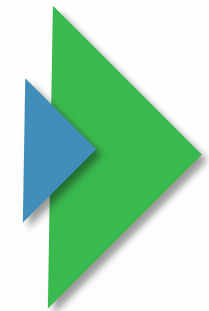
```
$ mkdir -p /home/user/.ssh  
$ chmod 700 ~/.ssh  
$ chmod 600 authorized_keys  
$ chown -R user:user ~/.ssh
```

Protect the .ssh directory, authorized_keys file, and ownership.

Exercise-1



- Create a user named **DevOps**
- Set the password for the **DevOps** user to **TechProEducation**
- Change the **DevOps** user's description to "**Best Cloud Engineer**"
- Switch to the **DevOps** user
- View the **DevOps** user's information
- Return to the previous user
- Delete the **DevOps** user along with their home directory



Group Management



Group Management



etc/group

Users can belong to several groups. Group membership is specified via the /etc/group file.

```
root@techproeducation:~# tail -1 /etc/group
linuxuser:x:1000:
root@techproeducation:~# tail -3 /etc/group
admin:x:115:
netdev:x:116:
linuxuser:x:1000:
root@techproeducation:~#
```

Group Management



groups

groups command is used to display a list of groups to which the user belongs.

groups [username]

```
root@techproeducation: ~# groups
root
root@techproeducation: ~# groups linuxuser
linuxuser : linuxuser
root@techproeducation: ~#
```

Group Management



groupadd

groupadd command is used to create a new group.

groupadd [groupname]

```
root@techproeducation: ~# groupadd linux
root@techproeducation: ~# groupadd aws
root@techproeducation: ~# groupadd devops
root@techproeducation: ~# tail -3 /etc/group
linux:x:1001:
aws:x:1002:
devops:x:1003:
root@techproeducation: ~# |
```

Group Management



usermod

You can change group membership with the usermod command.

usermod -a -G [groupname] [username]

- By default, if a group is not specified in the command, the usermod command will remove the user from any group!
- Using the -a (append) option prevents this behavior.

Admin Groups: Amazon Linux vs Ubuntu



Amazon Linux

```
$ sudo useradd -m developer
$ sudo usermod -aG wheel
developer
```

Administrative group: wheel

Ubuntu

```
$ sudo usermod -aG sudo
developer
```

Administrative group: sudo

Group Management



groupmod

groupmod command can be used to change the group name.

groupmod -n [newname] [oldname]

```
root@techproeducation: ~# tail -3 /etc/group
aws:x:1002:python
devops:x:1003:
python:x:1004:
root@techproeducation:~# groupmod -n java python
root@techproeducation:~# tail -3 /etc/group
aws:x:1002:python
devops:x:1003:
java:x:1004:
root@techproeducation:~# |
```

Group Management



groupdel

groupdel command is used to delete a group.

groupdel [groupname]

```
root@techproeducation: ~# tail -3 /etc/group
aws:x:1002:python
devops:x:1003:
java:x:1004:
root@techproeducation: ~# groupdel devops
root@techproeducation: ~# tail -3 /etc/group
linux:x:1001:
aws:x:1002:python
java:x:1004:
root@techproeducation: ~#
```

Group Management



gpasswd

With the **gpasswd** command, you can add a user to a group or remove a user from a group.

In the example below:

Using **gpasswd -a**, we add **john** to the **aws** group.

Using **gpasswd -d**, we remove **walter** from the **aws** group.

gpasswd -[option] [username] [groupname]

```
root@techproeducation:~#  
root@techproeducation:~# useradd john  
root@techproeducation:~# useradd walter  
root@techproeducation:~# gpasswd -a john aws  
Adding user john to group aws  
root@techproeducation:~# gpasswd -d walter aws  
Removing user walter from group aws  
gpasswd: user 'walter' is not a member of 'aws'  
root@techproeducation:~# gpasswd -a walter aws  
Adding user walter to group aws  
root@techproeducation:~# gpasswd -d walter aws  
Removing user walter from group aws  
root@techproeducation:~#
```

DevOps Group Strategy



Common groups

- developers
- admins
- docker
- aws

Commands

```
$ groupadd  
$ usermod -aG group user  
$ newgrp group
```

Exercise-2



1. Create a user named DevOps
2. Create two groups named Cloud and AWS
3. Add the DevOps user to the Cloud group
4. View the groups that the DevOps user belongs to
5. Add the DevOps user to the AWS group
6. Remove the DevOps user from the Cloud group
7. Delete the Cloud group
8. Rename the AWS group to AWS-CLOUD
9. View the groups that the DevOps user belongs to

Exercise-3



1. Create users: Jason, Bruce, Victor, Mark, Jack, Tyler, Tomy, Edward, Eric
2. Create groups: Asia, Europe, America, Africa
3. Add users to the Asia group: Jason, Bruce, Victor
4. Add users to the Europe group: Jason, Mark, Jack, Tyler
5. Add users to the America group: Jason, Tomy, Edward
6. Add users to the Africa group: Jason, Bruce, Mark, Edward, Jack, Tyler, Eric
7. View the groups that Jason belongs to
8. View the groups that Edward belongs to
9. Remove Bruce from the Africa group
10. Remove Jason from all groups
11. Delete the Europe group
12. Rename the Africa group to Australia

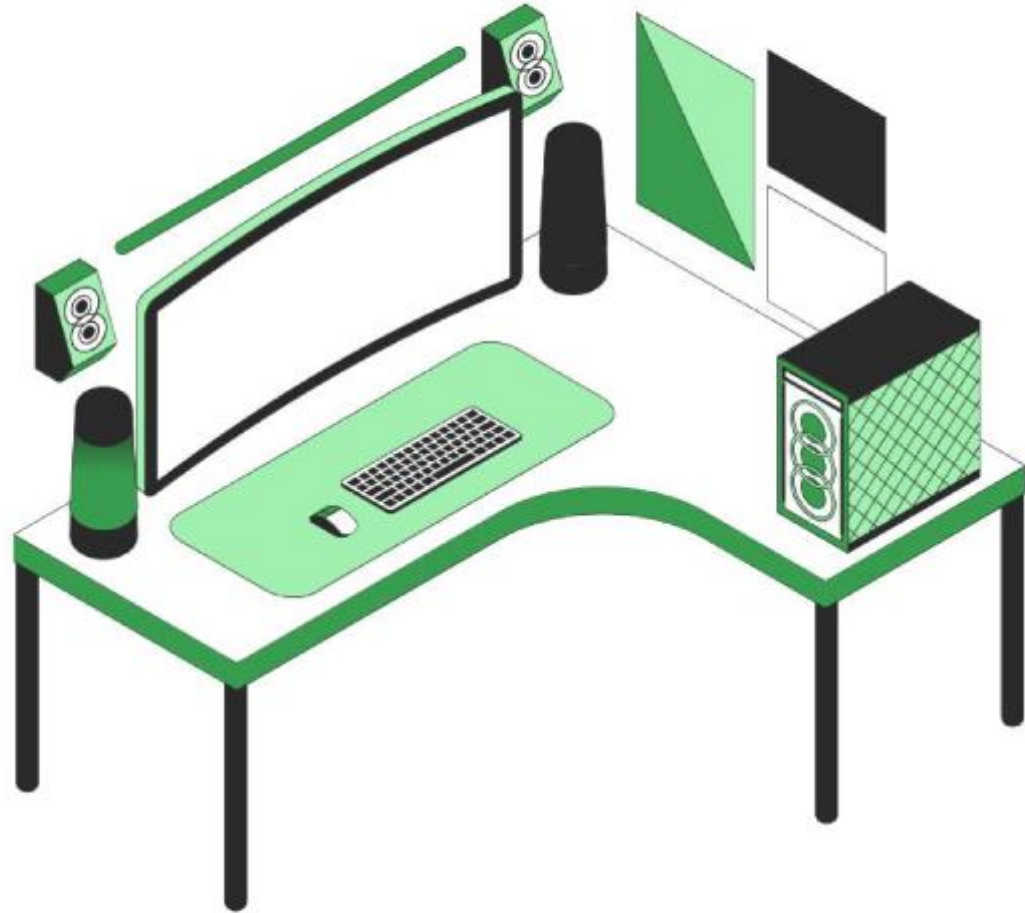
Session 6 — Key Takeaways



Key Skills Covered

- Linux users and groups
- sudo and privilege management
- Password and authentication concepts
- SSH access for EC2 users
- DevOps team-based group management

Practical Goal: Securely manage Linux servers in AWS environments



Do you have any questions?

Send it to us! We hope you learned something new.